

Technical Report: Computational Design of De Novo Protein Binders for RBX-1

Project Title: Strategic Truncation and Multi-Model Design of RBX-1 Binders

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Target Protein: RBX-1 (E3 Ubiquitin Ligase)

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1. Abstract

This report explains how we used computer tools to design new protein binders for RBX-1. RBX-1 is not an easy target because some parts of it are flexible and do not have a fixed structure.

To solve this, we removed those flexible parts and focused only on the stable regions. We worked with both the N-terminal and C-terminal parts of the protein using PDB ID: 3DQV.

We used two main tools: BindCraft (with soluble MPNN) and BoltzGen. These helped us create different binder designs that are more stable and more likely to bind well.

2. Target Analysis and Preparation

2.1 Intrinsically Disordered Region (IDR)

Truncation

RBX-1 has flexible regions that can make protein design harder. These regions do not have a clear structure.

To improve our results, we removed these flexible parts. This allowed our design tools to focus only on the stable parts of the protein where binding is more likely to happen.

2.2 Domain Specification

We used PDB ID: 3DQV as our main structure.

We focused on two parts:

Target A (N-Terminal): The structured N-terminal part of Chain A

Target B (C-Terminal): The structured C-terminal part of Chain A after removing

flexible regions

3. Design Methodology

3.1 N-Terminal Design: BindCraft with Soluble MPNN

For the N-terminal part, we used BindCraft on Tamarind Bio.

Scaffold Length: We set the binder length between 100 to 250 amino acids

Design Strategy:

Some designs were done without setting hotspots or residues

Other designs were done with specific hotspots or residues to guide binding

Optimization Method:

Instead of using AlphaFold to refold, we used soluble MPNN.

This helped improve:

Solubility

Stability

Better interaction at the binding surface

3.2 C-Terminal Design: BoltzGen

For the C-terminal part, we used BoltzGen on Tamarind Bio with default settings.

Chain Used: Chain A

Sequence Length: Between 100 to 250 amino acids

Design Strategy:

Some binders were designed without hotspots or residues

Others were designed with hotspots or residues

This gave us different types of binder structures to compare.

4. Design Selection and Validation

We checked all the designs using standard scores from the models.

We selected the best ones based on:

Structural Quality: Good pLDDT scores (shows stable structure)

Binding Confidence: Good iPAE and pTM scores (shows strong interaction)

Practical Use: Designs that are more likely to be soluble and easy to work with

(All detailed scores are in the CSV file.)

5. Conclusion

By removing flexible regions and using two different design tools, we created a set of RBX-1 binders that are more stable and likely to bind well.

Using both hotspot and non-hotspot designs also gave us more variety, which increases our chances of success in future testing.

These designs can help in studying or controlling RBX-1 activity through protein binding.